Valentin NOËL, PhD

Research Scientist and Research Programme Lead | Interpretability and AI Safety

val.noel@proton.me Paris, France linkedin.com/in/valnoelphd

Google Scholar github.com/vcnoel vcnoel.github.io

I work on understanding what happens inside large language models and on turning that understanding into things people can use. I lead research programmes across several entities of a consulting group, and I have shipped a compliance product built on that research. My published work covers reasoning, sycophancy, hallucination detection and the safety of agent pipelines.

Experience

Research Scientist — Devoteam Paris, France

Research programmes across the group. Interpretability, AI safety, applied LLM products. *Oct. 2024 – Present*

- Lead research programmes that run across several Devoteam entities, setting the research direction and coordinating the teams that deliver it. Supervise 8 interns and work with a consulting organisation of around ninety people.
- Designed and shipped the **Compliance Checker**, a product now available as a group asset for European AI Act compliance. It applies our research on prompt injection and hallucination detection to verify that generated output is grounded in the documents it cites and has not been steered by injected instructions.
- Built **HalluGraph**, a knowledge graph provenance verifier for legal retrieval augmented generation, running as an internal microservice. It reaches an AUC of **0.874** at detecting evidence that does not support the answer. The cases it cannot detect were measured and documented before anyone integrated it. Best Presentation Award, AAAI 2026.
- Showed that sycophancy is often carried by a single layer and can be suppressed by a closed-form, reversible edit to the weights, from **39.5% to 14.8%** on a benchmark of a thousand items. The edit costs accuracy on arithmetic reasoning, and that cost was measured rather than assumed. No retraining and no inference overhead.
- Developed a graph signal processing framework that locates load-bearing attention circuits from weights alone, giving a reasoning probe that needs no training and verifies valid reasoning at **95.6%** accuracy. Published at ICML 2026, main track.
- Measured how errors spread through agent pipelines. One wrong step left in an agent's context raises the failure rate of later, logically unrelated steps by about seven percentage points, and a stronger model is no more resistant than a weaker one. Built spectral kill switches that detect contamination before the next call.
- Validated these methods across five model families, from one to fourteen billion parameters, and published two preprints on the conditions under which such measurements stop being valid.

PhD Researcher — ENS Paris-Saclay & CentraleSupélec Paris, France

Deep Learning, Inverse Problems & Bayesian Methods — Supervisors: T. Rodet & D. Lesselier *Oct. 2021 – Sept. 2024*

- Bayesian deep learning for uncertainty quantification in inverse problems combining electromagnetic and ultrasound data. Published in *IEEE Transactions on Computational Imaging* and the *International Journal of Applied Electromagnetics and Mechanics*.
- Transfer learning strategies that treat reconstruction as the recovery of latent structure. The same framing underlies the circuit analysis work I do now.
- Trained and validated physics informed models on national supercomputers under SLURM, running distributed pipelines over high-dimensional data.

Grants & Funding

Silico Research Grants — Goodfire **USD 18,000** total (compute credits)

Two competitive individual awards. Spectral methods for open-weight model security. *2026*

- **USD 15,000** and **USD 3,000**, awarded separately. The programme covers tamper resistance of open weights, backdoor screening from weights alone, and checking a reward model for exploitable shortcuts before it is trained against.
- Two arXiv preprints were produced under the first award.

Selected Publications

V. Noël — Geometry of Reason: Spectral Signatures of Valid Mathematical Reasoning 2026

ICML 2026 (main track)

V. Noël*, R. Franzone*, P. Wang, P. Torr, F. J. Fehr — Building Reward Models for Grounded Legal Reasoning 2026

ICML 2026, AI for Law Workshop [*Equal contribution]

V. Noël — Sycophancy Is Often a Single-Layer Phenomenon: Spectral Diagnosis and Training-Free Weight Surgery in LLMs 2026

ICML 2026, MechInterp Workshop

V. Noël, K. Healy, V. Madathil, B. Srinivasan — Does the Optimal Hallucination Detector for Agentic Tool Calls Depend on Model Scale? 2026

ICML 2026, AIWILD Workshop

V. Noël — Spectral Archaeology: The Causal Topology of Model Evolution 2026
ACL 2026, Computational Developmental Linguistics Workshop

V. Noël — Catching Contamination Before Generation: Spectral Kill Switches for Agents 2026
AAAI 2026, TrustAgent Workshop

V. Noël — HalluGraph: Auditable Hallucination Detection for Legal RAG Systems via Knowledge Graph Alignment 2026
AAAI 2026, AILaw Bridge [Best Presentation Award]

V. Noël, T. Rodet, D. Lesselier — Electromagnetic Breast Imaging and Uncertainty Quantification with Bayesian Neural Networks 2023
PIERS 2023 [Best Student Paper Award]

Full list available on Google Scholar.

Preprints & Under Review

V. Noël — A Probe Direction Is a Property of Its Prompt 2026
arXiv:2608.13329 — under review. How much of an evaluation-awareness probe is the prompt wrapper rather than the model.

V. Noël — Where You Measure Decides What You Measure 2026
arXiv:2608.13337 — under review. A hidden degree of freedom in sparse-autoencoder ablation studies.

V. Noël — An Error in the Context Is a Demonstration: Measuring Error Contagion in Small-Model Agents 2026
Under review. Controlled measurement of self-injection across nine models and five families.

Education

PhD in Deep Learning — ENS Paris-Saclay and CentraleSupélec Paris, France
Supervisors: Thomas Rodet & Dominique Lesselier. Defended Oct. 2024. *Oct. 2021 – Sept. 2024*

MSc in Control and Robotics (Signal Processing & AI) — École Centrale de Nantes Nantes, France
Graduated 2nd in class. Focus: AI, inverse problems, signal processing. *Sept. 2019 – July 2021*

Awards

Best Presentation Award — AAAI 2026 Singapore
Association for the Advancement of Artificial Intelligence, AILaw Bridge *2026*

Best Student Paper Award — PIERS 2023 Prague, Czechia
PhotonIcs & Electromagnetics Research Symposium *2023*

Invited Talks

Invited speaker, AI and Mathematics Residency, International Centre for Mathematical Sciences, Edinburgh, 2026.

Speaker, NVIDIA GPU Technology Conference (GTC), Paris, 2025.

Guest lecturer, University of Electronic Science and Technology of China, Chengdu, 2025.

Guest talk, École Supérieure d'Ingénieurs Léonard de Vinci, Paris, 2025.

Skills

Interpretability & safety: Mechanistic interpretability, activation patching and steering, circuit analysis, sparse autoencoders, probing classifiers, attention head decomposition, graph signal processing, spectral methods, causal inference, red-teaming, behavioural evaluation harnesses (lm-eval), sycophancy and faithfulness benchmarks.

Machine learning: PyTorch, JAX, Hugging Face Transformers, PEFT/LoRA, CUDA/cuPy, DeepSpeed (DDP/FSDP/ZeRO), Hugging Face Accelerate, SLURM, Bayesian neural networks, inverse problems.

Programming & data: Python, C/C++, Julia, Matlab, SQL, PyTorch Geometric, DGL, NetworkX, Neo4j, FAISS, Milvus, pandas, Polars, NumPy.

Statistics: Bootstrap confidence intervals, cluster bootstrap, FDR correction, effect sizes, pre-registered analysis, power analysis.

Languages: French (native), English (C2), German (B2–C1).

Open Source, Teaching & Service

Lead Maintainer — `spectral-trust` (PyPI) 2025 – Present
Open-source transformer interpretability library — Python, PyTorch
- Real-time spectral graph diagnostics for LLMs: logical fallacy detection, connectivity-collapse monitoring, circuit-level trust metrics.

Lecturer — AI, Signal & Image Processing, Algorithms — ENS Paris-Saclay / Université Paris-Saclay Gif-sur-Yvette, France
Master's in Electronics, Electrical Energy and Automation *2021 – 2024*
- Three courses over three years, and supervision of dual-degree research interns.

Steering Committee Member — ESIGELEC-IRSEEM France
NMD ILTN Master's Programme — curriculum review, industry and alumni feedback *2026 – Present*

Member — Oxford Artificial Intelligence Society (OxAI) Oxford, UK
Led experimental design for contextual reward-modelling evaluation *2026 – Present*